

CAREFLOW DEMO GUIDELINE: HYPERTENSION IN ADULTS

Document code: CF-GL-HTN-02. Version 2. Owner: Department of Cardiology.

SYNTHETIC DEMONSTRATION DOCUMENT. Written for the CareFlow AI portfolio project; fictional and not real clinical guidance.

1. Scope

This guideline covers the diagnosis and outpatient management of primary hypertension in adults aged 18 and over. Hypertension in pregnancy and in children follows separate protocols.

2. Blood Pressure Measurement

Measure blood pressure after the patient has been seated for 5 minutes with the arm supported at heart level, using a correctly sized cuff. Record the average of at least two readings taken one to two minutes apart. Confirm a new diagnosis with ambulatory blood pressure monitoring (ABPM) or home blood pressure monitoring (HBPM) before starting long-term treatment, unless blood pressure is 180/110 mmHg or higher.

3. Classification

Stage 1 hypertension is a clinic blood pressure of 130 to 139 mmHg systolic or 80 to 89 mmHg diastolic. Stage 2 hypertension is 140/90 mmHg or higher. A reading of 180/120 mmHg or higher is a hypertensive crisis.

4. Treatment Targets

The target for most adults is below 130/80 mmHg. For adults over 80 years or with frailty, a target below 140/90 mmHg is acceptable. Patients with diabetes or chronic kidney disease should aim for below 130/80 mmHg when tolerated.

5. Drug Treatment

5.1 First-line agents

First-line options are an ACE inhibitor (for example lisinopril), an angiotensin receptor blocker (for example losartan), a dihydropyridine calcium channel blocker (for example amlodipine) or a thiazide-type diuretic (for example hydrochlorothiazide). Do not combine an ACE inhibitor with an angiotensin receptor blocker.

5.2 Combination therapy

Start two first-line drugs from different classes when blood pressure is more than 20/10 mmHg above target. A single-pill combination improves adherence.

5.3 Chronic kidney disease and albuminuria

In patients with albuminuria (UACR above 30 mg/g) an ACE inhibitor or angiotensin receptor blocker is preferred and should be titrated to the maximum tolerated dose.

5.4 Laboratory monitoring

Check serum potassium and creatinine 1 to 2 weeks after starting or increasing an ACE inhibitor, angiotensin receptor blocker or diuretic. A creatinine rise of up to 30% is acceptable; a larger rise or potassium above 5.5 mmol/L requires dose review.

6. Hypertensive Urgency and Emergency

A blood pressure of 180/120 mmHg or higher WITHOUT acute target-organ damage is hypertensive urgency and can usually be managed by restarting or intensifying oral therapy with review within 24 to 72 hours. The same reading WITH acute target-organ damage (chest pain, neurological deficit, acute kidney injury, pulmonary oedema) is a hypertensive emergency and requires emergency admission through the Emergency Admission Procedure.

7. Lifestyle

Advise sodium intake below 2 g per day, regular physical activity of at least 150 minutes per week, weight reduction when overweight, limiting alcohol and stopping smoking.

8. Follow-up

Review monthly after any treatment change until the target is reached, then every 3 to 6 months.